

Swiss cheese model in practice



**ROBUST
INTELLIGENCE**

AI Firewall



Guardrails

 **Meta AI**

**PurpleLlama,
LlamaGuard**



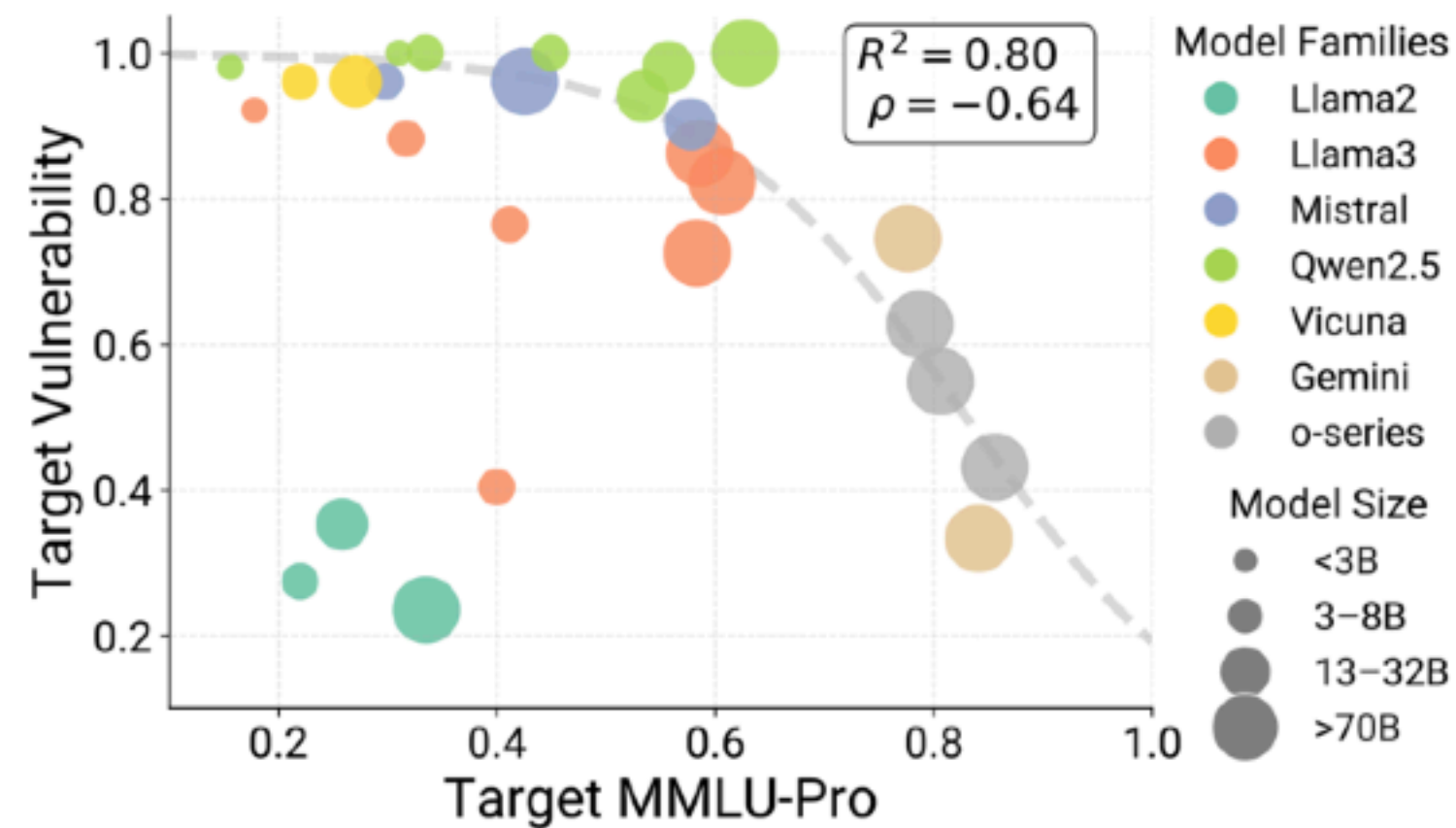
Cygnal



VirtueGuard

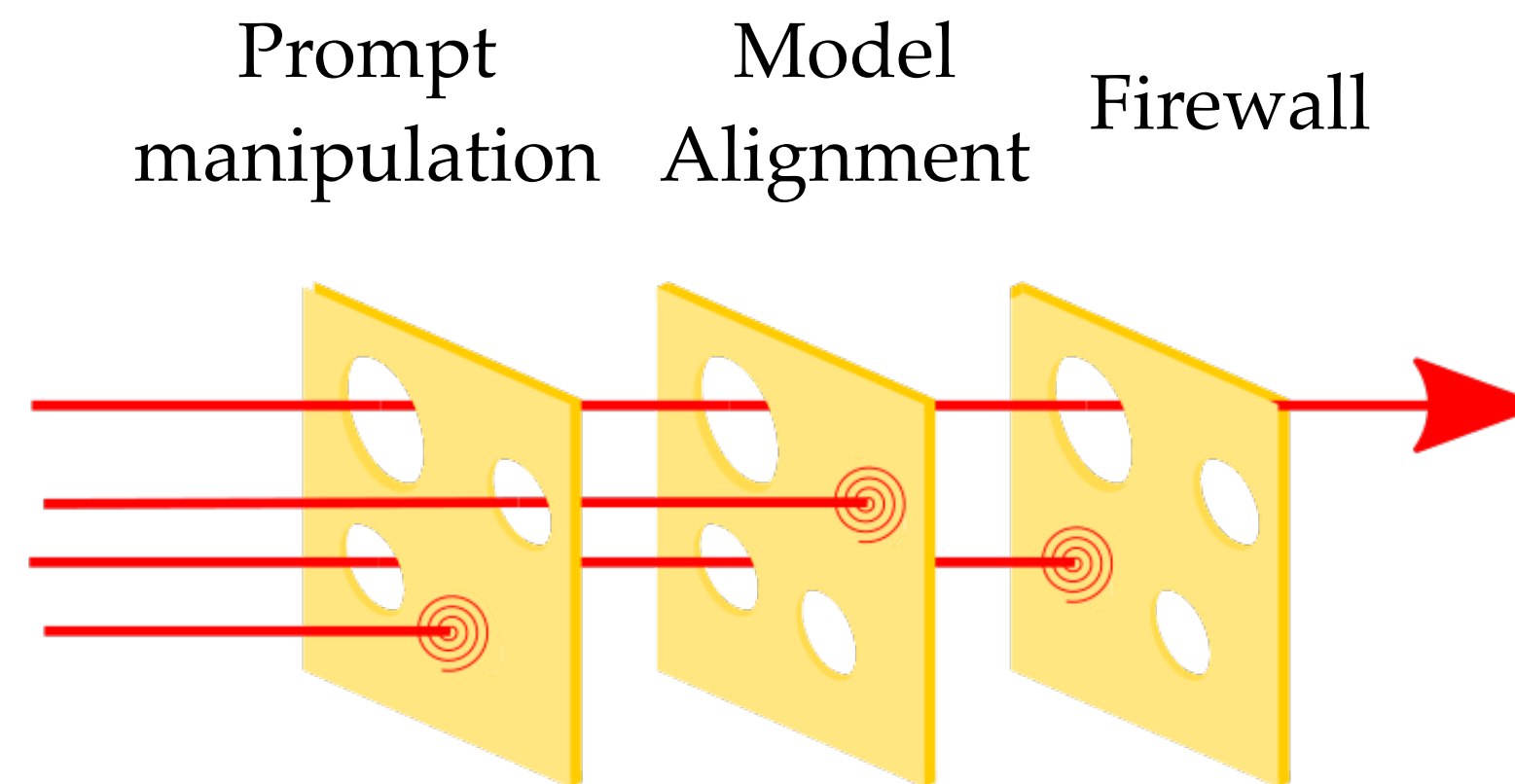
Open directions

There is no perfect method yet; lots of opportunities for future research.



Is bigger always safer?

(Scaling laws; Panfilov et al., 2025).



Is post-hoc mitigation required for safety?

(STACK; McKensize et al., 2025).



Are jailbroken outputs useful?

(Jailbreak tax; Nikolic et al., 2025).